# FlavorMap

## แผนที่รสชาติ

*Project Summary · Pillar 2 of the Future Leader Plan · ISB Bangkok Grade 10 · Data Science Portfolio*

## What the project is

> *"Thailand has 77 provinces and 77 distinct culinary traditions. FlavorMap is the first computational study to map them — using ingredient network analysis and geographic clustering to answer whether food encodes cultural geography."*

Thai cuisine is described internationally — and even within Thailand — as a single entity. But anyone who has eaten across the country knows this is wrong. Northern Thai food uses galangal and dried chilies almost absent from Southern cooking. Isaan cuisine uses pla ra (fermented freshwater fish) that Central Thai cooks rarely touch. Southern curries carry coconut milk and fresh turmeric reflecting four centuries of maritime spice trade. FlavorMap asks: can data science see what the dominant food narrative erases?

### Four technical components

| Component | What it does | Key method |
|---|---|---|
| Ingredient network analysis | Builds a graph where nodes are ingredients and edges represent co-occurrence in recipes. Community detection finds the 'flavor families' of Thai cooking. | NetworkX + Louvain community detection |
| Geographic clustering | Represents each province as an ingredient vector. Maps culinary similarity across all 77 provinces. Tests whether similar cuisines cluster geographically. | UMAP + K-means + Moran's I spatial autocorrelation + Mantel test |
| Province-of-origin classifier | ML model predicts which province a dish comes from based only on ingredients. The confusion matrix reveals which provinces share culinary traditions. | Random Forest + feature importance analysis |
| Interactive visualization | Live map at flavormap.org — click any province to see its most distinctive ingredients, culinary similarity to neighbors, and sample recipes. | Mapbox GL JS + Sigma.js + Next.js |

Data comes from two sources: web scraping of six major Thai recipe databases (~1,500 recipes) and structured fieldwork interviews with 20–30 home cooks from provinces across Thailand (~500 additional recipes with high-confidence province attribution). The fieldwork is irreplaceable — the most distinctive provincial recipes exist only in the kitchens of elderly cooks who never posted them online.

### Academic output

- **Research paper:** "Ingredient networks as cultural geography: a computational analysis of Thai regional cuisine variation across 77 provinces" — first author, co-authored with Chulalongkorn University faculty. Target: Digital Humanities 2026 (DH2026). Fallback: arXiv preprint Month 11.
- **Public dataset:** OpenFlavorTH released on HuggingFace — the first open-access structured dataset of Thai regional cuisine, including canonical ingredient dictionary and network analysis data. Verifiable public download count.

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## What impact this project creates

| For whom | Impact |
|---|---|
| Thai home cooks and communities | Recipes of elderly rural home cooks are disappearing as families urbanize. FlavorMap documents them — not as museum pieces but as a living dataset that proves provincial cooking traditions are distinct and culturally significant. For a grandmother in Surin whose pla ra recipe has never been written down, FlavorMap is the first system to record and contextualize her knowledge. |
| Food researchers and digital humanists | No structured, province-level Thai recipe dataset exists publicly. Every researcher studying Thai culinary geography must currently build their own data from scratch. OpenFlavorTH eliminates that friction permanently. The canonical ingredient dictionary — mapping hundreds of variant Thai names to standardized forms — is itself a standalone research contribution. |

| For whom | Impact |
|---|---|
| Food industry and tourism sector | Province-level ingredient distinctiveness data has direct commercial applications for regional product development, localized menus, and culinary tourism itineraries. FlavorMap makes quantitatively what was previously only anecdotal knowledge. |
| The broader 'culture is infrastructure' thesis | FlavorMap provides quantitative evidence that 'Thai food' as a unified brand erases real cultural diversity — 77 distinct culinary traditions exist where one narrative has been marketed. This is the same argument TunDee makes about scholarships. Together, the two projects are two empirical demonstrations of one claim: information infrastructure systematically erases communities farthest from the center. |

## Why FlavorMap is the best project to work on after TunDee for university applications

- **Demonstrates skills TunDee cannot**

TunDee covers supervised ML, Thai NLP, fairness-aware ML, causal inference, and full-stack deployment. FlavorMap covers graph theory, network analysis, unsupervised learning, geospatial data science, interactive visualization, ethnographic fieldwork, and dataset curation. An admissions reader sees two genuinely distinct technical paradigms — not one skill applied twice.

- **Shifts from product to research**

TunDee is product-first: a deployed platform with users. FlavorMap is research-first: a finding, a paper, and a public dataset. Having both modes shows methodological range that CMU iSchool, Cornell IS, UMich SI, and Berkeley CDSS explicitly train for and rarely see demonstrated at the high school level.

- **Authentic personal connection**

She loves cooking. She grew up eating Thai food from different regional backgrounds. Her grandmother's cooking tastes different from Bangkok restaurant versions. This is not a project invented for admissions — it is a question she is genuinely qualified to ask and curious about. Admissions readers distinguish performed from authentic interest.

- **The essay moment is irreplaceable**

The grandmother's tom yum — recognizing her family recipe as not personal nostalgia but centuries of Isaan trade routes encoded in a soup — is a moment no other applicant can write. Specificity is the essay's most valuable quality, and FlavorMap provides specificity entirely hers.

- **Creates a coherent intellectual identity**

Without FlavorMap, TunDee is a strong social impact project. With FlavorMap, the two together become an argument: information infrastructure systematically erases communities farthest from the center, demonstrated twice, with two different datasets, using two different methods. This is an intellectual identity, not a portfolio.

- **The dataset is permanent and verifiable**

TunDee's impact requires trust in user count claims. The OpenFlavorTH dataset on HuggingFace has a public download count any admissions reader verifies with one click. It cannot be inflated. It is the most credible impact metric in the application.

- **Generates revenue that funds TunDee**

FlavorMap data licensing to Thai food companies and tourism organizations creates a revenue stream that funds TunDee's free tier for rural students. The GSEA social enterprise pitch frames both as one self-sustaining civic data ecosystem — not two separate projects.

**The one-paragraph version.** FlavorMap uses ingredient network analysis and geographic clustering to build the first computational map of Thai regional cuisine — analyzing 2,200+ recipes across all 77 Thai provinces to answer whether food encodes cultural geography. The interactive visualization at flavormap.org, the research paper (targeting Digital Humanities 2026), and the OpenFlavorTH public dataset on HuggingFace are the academic outputs. The project extends her thesis from TunDee — that information infrastructure systematically erases communities farthest from the center — into a new domain, using new technical methods, with a finding that is genuinely hers: her grandmother's recipe is not sentiment. It is data.